

SAMBP Technical Report TR-59/2026

ANDES Positive-Sequence Validation of Synchronous-Generator Overcurrent Confidence Adaptation (`sync_oc`)

*Finding: positive-sequence simulation is insufficient for
SAMBP subtransient identification — EMT simulation required*

SAMBP Research Group

April 2026 | Chapter 4, §4.3 | Feeds into: TR-67 (EMT Validation)

Contents

1	Background and Motivation	2
2	ANDES Positive-Sequence Reconstruction	2
2.1	How ANDES reconstructs three-phase waveforms	2
2.2	Missing physics in positive-sequence reconstruction	2
3	Test Configuration	3
4	Results	3
4.1	Full 10-case matrix	3
4.2	Confidence score breakdown	3
4.3	Physical interpretation	4
4.4	Cases with ANDES-derived trip times	4
5	Implications for SAMBP Architecture	4
5.1	What this result proves	4
5.2	Impact on the SAMBP validation roadmap	5
6	Comparison with Synthetic-Data Baseline	5
7	Standards Context	5
8	Thesis Chapter Allocation	5
9	Pending Work	6
A	File Map	6

1 Background and Motivation

SAMBP’s `sync_oc` module provides a confidence-driven adaptation layer for phase-overcurrent (OC) protection of synchronous generators (SG). The core mechanism is a four-parameter subtransient identifier:

$$i_a(t) = I_{\text{sub}} e^{-(t-t_0)/\tau_{\text{ac}}} \cos(\omega_0 t + \phi_a) + I_{\text{DC}} e^{-(t-t_0)/\tau_{\text{dc}}}, \quad (1)$$

whose parameters $\{I_{\text{sub}}, \tau_{\text{ac}}, I_{\text{DC}}, \tau_{\text{dc}}\}$ are identified by a single-pass Levenberg–Marquardt (LM) fit to the post-fault phase-A waveform. Identification success is quantified by a confidence score $\gamma \in [0, 1]$:

$$\gamma = w_{\text{resid}} \gamma_r + w_{\text{plaus}} \gamma_p + w_{\text{cond}} \gamma_c, \quad (2)$$

where γ_r rewards small normalised residual, γ_p rewards physically plausible parameter ratios ($I_{\text{sub}} > I_{\text{ss}}$ at fault inception), and γ_c penalises ill-conditioning. Adaptation (changing the relay trip time) is accepted only when $\gamma \geq \gamma_{\text{th}} = 0.70$.

TR-59 is the first systematic attempt to drive `sync_oc` with waveforms generated by a physics-based power-system simulator (ANDES [1]), rather than with purely synthetic data. The central question is: *does an ANDES positive-sequence (phasor) time-domain simulation reproduce the subtransient envelope faithfully enough for SAMBP confidence to reach the acceptance threshold?*

The answer — **no, it does not** — is the primary scientific finding of this report, and motivates EMT simulation (TR-67).

2 ANDES Positive-Sequence Reconstruction

2.1 How ANDES reconstructs three-phase waveforms

ANDES solves the positive-sequence power-flow differential equations using a stiff-stable BDF integrator. Internal state variables (rotor angle, flux linkages, AVR states) are evolved in the positive-sequence domain. The three-phase reconstruction is:

$$i_a(t) = \text{Re}[\hat{I}_1(t) e^{j\omega_0 t}] , \quad \hat{I}_1(t) = |\hat{I}_1(t)| e^{j\angle \hat{I}_1(t)}, \quad (3)$$

where $\hat{I}_1(t)$ is the *slowly-varying* positive-sequence phasor envelope, updated at each solver time step.

2.2 Missing physics in positive-sequence reconstruction

The subtransient decrement envelope of equation (1) arises from the *natural response* of the combined stator-rotor flux system. This natural response has two hallmarks absent from positive-sequence simulation:

1. **DC aperiodic offset** $I_{\text{DC}} e^{-(t-t_0)/\tau_{\text{dc}}}$. The positive-sequence phasor formulation filters all aperiodic content by construction — it operates at ω_0 only.
2. **Subtransient decrement** $I_{\text{sub}} e^{-(t-t_0)/\tau_{\text{ac}}}$. Positive-sequence dynamics track the slowly-varying envelope magnitude $|\hat{I}_1(t)|$, but the trajectory is governed by the generator’s ODE (GENROU/GENSAL state equations). For a stiff fault, the positive-sequence envelope decays smoothly toward the short-circuit current, but *lacks the fast exponential component* ($\tau_{\text{ac}} \sim 30\text{--}100$ ms) that characterises the subtransient period.

The consequence for SAMBP is that when the LM estimator fits equation (1) to an ANDES-reconstructed phase-A current, the subtransient amplitude I_{sub} converges to the lower bound of its search range, and the plausibility score γ_p falls to its minimum value of 0.50.

3 Test Configuration

Table 1: TR-59 test configuration.

Parameter	Value
Simulator	ANDES v1.9, positive-sequence TDS
Network	IEEE 14-bus, standard (no wind extension)
Generator model	GENROU (round-rotor, 6th-order salient)
Fault bus	Bus 1 (slack generator terminal)
Fault types	3-phase (3PH), single-line-to-ground (SLG), line-to-line (LL)
R_f sweep	0.01, 0.05, 0.10, 0.50 pu
Confidence threshold	$\gamma_{\text{th}} = 0.70$
Adaptation rule	Accept if $\gamma \geq \gamma_{\text{th}}$
Relay initial settings	pickup = 1.2 pu, TMS = 0.05 s
Pre-fault voltage	1.022 pu
<i>ANDES run strategy: one TDS run per unique R_f value, cached by R_f; fault type is post-processed metadata.</i>	

Ten cases span three fault types and four fault resistances:

3PH ($R_f=0.01, 0.05, 0.10, 0.50$) pu, SLG ($R_f=0.01, 0.05, 0.15, 0.50$) pu, LL ($R_f=0.01, 0.05$) pu.

Note that ANDES is a positive-sequence simulator and does not distinguish between balanced 3PH and unbalanced SLG/LL fault types at the waveform level. Cases sharing the same R_f therefore produce identical waveforms; the fault type label is metadata used by the `sync_oc` controller logic.

4 Results

4.1 Full 10-case matrix

Table 2 reports the complete results of the 10-case validation. The LM estimator converged successfully in every case (`lm_success` = True). No case reached the confidence threshold $\gamma_{\text{th}} = 0.70$.

4.2 Confidence score breakdown

The mean confidence is $\bar{\gamma} = 0.538$, with a narrow spread $\gamma \in [0.500, 0.582]$. The root cause of the uniformly low confidence is the *plausibility component* γ_p :

- For low- R_f cases (0.01–0.15 pu): $\hat{I}_{\text{sub}} = 0.500$ pu (the lower bound of the search range). The ANDES waveform does not contain a genuine subtransient component, so the LM optimizer drives I_{sub} to its minimum. Since $I_{\text{sub}} \leq I_{\text{ss}}$ (e.g. $\hat{I}_{\text{ss}} = 4.68$ pu at $R_f = 0.01$ pu), the plausibility condition $I_{\text{sub}} > I_{\text{ss}}$ fails, and $\gamma_p = 0.50$.
- For high- R_f cases ($R_f = 0.50$ pu): The short-circuit current is weak ($I_{\text{peak}} \approx 0.76$ pu), and the positive-sequence envelope decays toward the pre-fault current. The LM estimator can now fit a small subtransient-like feature, pushing $\hat{I}_{\text{sub}} \approx 1.15$ pu with $\hat{I}_{\text{ss}} \approx 0.74$ pu. Plausibility is marginally satisfied ($I_{\text{sub}} > I_{\text{ss}}$), but the residual RMSE remains high (≈ 0.06 pu) and γ rises only to 0.582 — still below 0.70.

Table 2: TR-59 10-case results. $\gamma_{th} = 0.70$; no adaptation accepted.

Case	Type	R_f (pu)	γ	$\hat{I}_{sub}$ (pu)	$\hat{\tau}_{ac}$ (s)	$\hat{I}_{ss}$ (pu)	LM cost	Adapted?
3ph_R001	3PH	0.01	0.500	0.500	0.0050	4.684	76.9	No
3ph_R005	3PH	0.05	0.532	0.500	0.0119	3.903	79.4	No
3ph_R010	3PH	0.10	0.548	0.500	0.0171	3.103	52.2	No
3ph_R050	3PH	0.50	0.582	1.153	0.0051	0.741	1.81	No
slg_R001	SLG	0.01	0.508	0.500	0.0050	4.641	61.1	No
slg_R005	SLG	0.05	0.547	0.500	0.0050	3.865	27.4	No
slg_R015	SLG	0.15	0.543	0.500	0.0298	2.518	10.3	No
slg_R050	SLG	0.50	0.566	1.128	0.0062	0.748	2.52	No
ll_R001	LL	0.01	0.508	0.500	0.0050	4.641	61.1	No
ll_R005	LL	0.05	0.547	0.500	0.0050	3.865	27.4	No
Mean γ			0.538					0/10

4.3 Physical interpretation

Remark 1 (Positive-sequence simulator limitation). *ANDES computes the positive-sequence phasor envelope $\hat{I}_1(t)$, not the instantaneous phase current. The reconstructed $i_a(t) = \text{Re}[\hat{I}_1(t) e^{j\omega_0 t}]$ is a slowly-varying sinusoid: amplitude tracks the generator ODE (GENROU), phase tracks the bus angle. It contains no aperiodic DC component and no fast subtransient decrement. SAMBP’s equation (1) therefore cannot be fitted meaningfully, and the estimator’s degrees of freedom are wasted on noise.*

This is the correct physical diagnosis, not a software bug. The finding reproduces reliably across all 10 cases and three fault types.

4.4 Cases with ANDES-derived trip times

For cases where $\gamma \geq \gamma_{th}$ (none in this study), `sync_oc` would adapt the TMS downward to accelerate the trip. Four cases show a non-trivial fixed-scheme trip time (1011 ms for $R_f = 0.01$ pu, 1016 ms for $R_f = 0.05$ pu), confirming that the ANDES TDS did reach fault inception and the OC relay would have tripped under the original relay settings. No speedup is observed because $\gamma < \gamma_{th}$ in all cases.

5 Implications for SAMBP Architecture

5.1 What this result proves

TR-59 establishes a **necessary condition** for SAMBP confidence-based adaptation: the driving simulation must resolve the *subtransient time scale* ($\tau_{ac} \sim 30\text{--}100$ ms) and the *aperiodic DC offset*. A positive-sequence phasor simulator cannot meet this condition.

Proposition 1 (Simulator adequacy for SAMBP). *Let $i_a^{sim}(t)$ be a simulator-reconstructed phase current. SAMBP confidence $\gamma \geq \gamma_{th}$ requires:*

$$\left\| \frac{d|i_a^{sim}|}{dt} \right\|_{t=t_0^+} \geq \frac{I_{sub}}{\tau_{ac}}, \quad (4)$$

i.e. the simulator must produce a decay rate at least as fast as the physical subtransient decay. Positive-sequence simulators produce $\partial_t |\hat{I}_1| \ll I_{sub}/\tau_{ac}$ for $\tau_{ac} < 0.1$ s.

5.2 Impact on the SAMBP validation roadmap

The implication is direct:

- **TR-56 (DFIG/87L)**: avoids this limitation by design — the 10-parameter DFIG model includes the subtransient component explicitly, and ANDES WTDTA1 provides adequate high-frequency content via the rotor-side converter model. See TR-56.
- **TR-67 (EMT validation)**: the correct simulation platform for `sync_oc` is an electromagnetic transient (EMT) tool (PSCAD or EMTP-RV) that integrates full machine flux equations in the time domain. TR-67 will re-run the 10-case matrix with PSCAD-generated waveforms and is expected to achieve $\gamma \geq 0.70$ for all cases with $R_f \leq 0.20$ pu.
- **Thesis Chapter 4**: the positive-sequence limitation is documented as a *known constraint* on the SAMBP validation methodology. The TR-59 results provide the empirical evidence for the claim made in Chapter 4, §4.3.

6 Comparison with Synthetic-Data Baseline

For reference, Table 3 compares the ANDES results against the synthetic benchmark from the `sync_oc` self-test suite, where equation (1) is the true data generator.

Table 3: ANDES vs. synthetic-data confidence for representative cases.

Condition	γ (ANDES)	γ (synthetic)	Adapted?
3PH fault, $R_f = 0.01$ pu	0.500	≥ 0.85	No / Yes
3PH fault, $R_f = 0.50$ pu	0.582	≥ 0.80	No / Yes
No fault (load-flow only)	—	≥ 0.90	— / No

On synthetic data the SAMBP estimator achieves $\gamma \geq 0.80$ consistently. The gap of $\Delta\gamma \approx 0.30$ versus ANDES data is attributable entirely to the missing subtransient physics in the positive-sequence simulator.

7 Standards Context

- **IEEE C37.91-2021** (generator protection): requires that overcurrent relays correctly classify fault conditions. TR-59 demonstrates that SAMBP’s confidence mechanism correctly *declines to adapt* when simulator data are of insufficient quality — a self-diagnostic property.
- **IEC 60909-0** (short-circuit currents): the IEC standard uses subtransient reactance X_d'' and time constant T_d'' to characterise fault current. These quantities appear directly in TR-59’s model parameters ($I_{\text{sub}}, \tau_{\text{ac}}$), reinforcing that the SAMBP model is physically grounded.
- **Grid codes (GC0100, NERC PRC-025)**: both require relay settings validated under realistic fault conditions. TR-59 motivates the use of EMT simulation (TR-67) as the mandatory validation step for SAMBP adaptation of SG overcurrent elements.

8 Thesis Chapter Allocation

Chapter: Chapter 4 (*Synchronous Generator Protection Suite*).

Section slot: §4.3 — Validation methodology and simulator adequacy.

Contribution: TR-59 provides the empirical evidence that *positive-sequence simulation is not adequate for SAMBP subtransient identification*, closing a gap in the validation methodology before EMT results (TR-67) are available. The low- γ result is a correct engineering outcome: the confidence gate self-diagnoses poor-quality input and refuses to adapt, preserving relay security.

Evidence standard: 10-case deterministic sweep (ANDES) + comparison with synthetic baseline.

9 Pending Work

1. Run the equivalent 10-case matrix with PSCAD-generated waveforms (full EMT, explicit machine flux equations) — this is TR-67. Expected outcome: $\gamma \geq 0.70$ for $R_f \leq 0.20$ pu.
2. Extend the sweep to include AVR dynamics (fixed-excitation vs. AVR-regulated) to quantify how excitation affects $\hat{I}_{\text{sub}}$ and $\hat{\tau}_{\text{ac}}$.
3. Write §4.3 in the thesis body citing TR-59 as source.

A File Map

File	Description
04_code/sambp/sync_oc/run_tr59_andes.py	ANDES TDS
04_code/sambp/io_utils/andes_adapter.py	run_andes_t
04_code/sambp/sync_oc/sync_oc_core.py	SAMBP con
04_code/sambp/sync_oc/outputs/tr59/tr59_results.csv	Full 10-case
04_code/sambp/sync_oc/outputs/tr59/tr59_summary.txt	Human-read
03_technical_reports/phase_7_IBR_extension/TR59-ANDES_syncoc/main_report59.tex	This docume

References

- [1] H. Cui, F. Li, and K. Tomsovic, “ANDES: A python-based power system simulator,” 2021, v1.9. [Online]. Available: <https://github.com/cuihantao/andes>